

N136AV (S/N: KONUK10295858) — Modem QDL Recovery Failure Diagnostic Report

Immfly Ground Operations / IT — Session: September 15, 2026

Field	Value
Machine	N136AV (S/N: KONUK10295858)
Affected component	Sierra Wireless EM7565 modem — stuck in QDL (download/boot-and-hold) mode
Initial symptom	No cellular connectivity; modem_fix.py skipped firmware upgrade due to known script logic bug
USB identification	1199:9090 (QDL/recovery mode) instead of normal operating mode 1199:9091
Root cause	Modem stuck in QDL mode; firmware recovery via qmi-firmware-update fails at protocol handshake ('HDLC trailing control character not found'), even after a full machine reboot
Comparison	Byte-for-byte identical failure signature to previously documented N060AV case
Recommended action	Physical modem replacement required — software-level recovery exhausted

Narrative Summary

N136AV had no cellular connectivity. The automated modem repair script correctly diagnosed old firmware (SWI9X50C_01.08.04.00) in MBIM mode, but its fix-application phase skipped the upgrade due to a known logic bug. Manual investigation revealed the modem was enumerating as **1199:9090** (QDL/download mode) instead of the normal **1199:9091**.

A firmware recovery attempt using `qmi-firmware-update -U -t /dev/ttyUSB3` was made with the fleet's standard firmware files. The tool validated the images successfully and the modem responded with 48 bytes during the QDL handshake, but the tool failed to parse the response: **'HDLC trailing control character not found'** — byte-for-byte identical to the N060AV failure signature.

A full machine reboot was performed to rule out a transient state. Post-reboot, the modem remained in QDL mode (1199:9090). Given the identical outcome to N060AV — where additional recovery attempts were all unsuccessful and one carried a risk of worsening the modem's state — no further attempts were made.

Troubleshooting Steps Performed

1. Automated fix script

modem_fix.py correctly detected old firmware and MBIM mode but skipped the upgrade due to the known script logic bug.

2. USB bus check

lsusb confirmed 1199:9090 (QDL mode). dmesg showed only ttyUSB3 via qcserial — no qmi_wwan or cdc-wdm interface.

3. Firmware recovery attempt

ModemManager stopped. qmi-firmware-update -U -t /dev/ttyUSB3 run with fleet firmware files. Failed: 'unsupported download protocol'.

4. Verbose diagnostic

Re-run with -v. Sahara handshake failed. QDL attempted: modem responded with 48 bytes but tool failed: 'HDLC trailing control character not found'. Identical to N060AV.

5. Full machine reboot

Power-off/power-on cycle performed. Post-reboot lsusb confirmed modem still 1199:9090.

6. Decision to stop

Given identical outcome to N060AV and demonstrated risk of worsening the modem state, no further recovery attempts were made.

Conclusion and Recommendation

N136AV's modem is stuck in QDL mode and cannot be recovered through the standard software recovery path. The failure is persistent across a full power cycle and is byte-for-byte identical to the N060AV case.

Recommended action: Escalate N136AV for physical modem replacement, following the same repair-order workflow used for N060AV.

Appendix: Key Command Output

```
lsusb (before and after reboot):
  Bus 001 Device 007: ID 1199:9090 Sierra Wireless, Inc.

qmi-firmware-update verbose (key excerpt):
[qfu-sahara-device] initializing sahara protocol...
[qfu-updater] sahara device creation failed: no sahara response received
[qfu-qdl-device] >> 7E:70:00:00:14:46:7E [7]
[qfu-qdl-device] << 01:00:00:00:30:00:00:00:02:00:00:00:01:00:00:00:00:04:00:00:
                    02:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:
                    00:00:00:00:00:00:00 [48]
[qfu-updater] qdl device creation failed: HDLC trailing control character not found
error: unsupported download protocol
```